

ActInf Textbook Group ~ Cohort 2 ~ Meeting 23 (Chapter 10, part 2)

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Hello and welcome. We're in cohort two of the Parr et al textbook group.

We are in our final discussion on chapter 10, the last of the topical discussions.

So we'll see where it goes. Then next week will be the final meeting of this cohort, where we'll have unrecorded feedback, discuss some project ideas, and so on. So this will be the last recording in this series for cohort two. And please go to future textbook groups to fill out the feedback form and or share the link to register for future cohorts. That being said, let us turn to chapter 10 and then go wherever people are interested. So Michael, maybe you can restate what area you want to look at, and then let's see if it relates to anything in 10.

Michael Schultz- Yeah, okay. So my question was related to priming effects. So priming is very much used paradigm in psychology, which assumes that the stimulus activates somehow a network of semantic related items or phonetic or somehow related items, which facilitates recognition of successive items and then according also action. And I was wondering how this is represented in active inference, because it's a kind of an action perception loop, and how the difference between more conscious or more higher level kinds of decisions are distinguished in this active inference framework. Whether we have a possibility to distinguish between these kinds of more priming action perception loops and more consciously activated action perceptions. Are there any papers? Is there any research that that somehow maps those different distinctions into the terminology that we have in active inference? Do you know?

Michael Schultz- Yep. Ali, any thoughts?

Ali Kishnicka

Ali Kishnicka

Nothing immediately comes to mind, but you go on, please. And then I'll think if I can add to it later.

Okay. One,

this one appears to at least address attentional priming in the context of motor behavior.

I'll add the link into the notes section.

But I think as we're coming to see, the framework is going to be minimally opinionated on complex, extremely heterogeneous cognitive phenomena like priming.

You can construct a cognitive model that embodies that through any number of ways.

Like you could imagine a nested perceptual model where the first, the higher level is what genre of movie we're watching, comedy or horror.

And then the stimuli are the sounds.

And then like hearing a laugh,

then updates the belief that it's a comedy, not a horror.

And then that makes certain stimuli more attended to in the future.

But that wouldn't be like an opinionated active inference framework describing how priming is approached,

but rather a way that you could use the kinds of techniques that we discussed in the textbook to write your own generative model that has some of the features that you'd like.

And I'm not exactly sure how they've done it in this motor paper all the way from 2011.

Okay.

Ali, anything?

Ali, anything?

Yeah, I just wanted to add that.

I'm not sure if it directly relates to this specific question or not, but at least I haven't seen much paper doing direct empirical experiments,

trying to, I mean, trying to observe or confirm some of the ideas of active inference framework.

In other words, most of the papers in the literature are mostly concerned with modeling the existing experimental data or empirical data.

And, but there are some important works that's being done, not directly trying to prove or confirm the active inference framework,

but they have done some research in other areas, but they've used active inference in the modeling of the phenomena being observed or studied.

So in this case also, I vaguely remember seeing some related studies that's being done and ultimately modeled with active inference framework,

but I'm not sure if there's anything specifically putting active inference claims as hypothesis and trying to confirm those hypothesis.

So, as you said, active inference is not, cannot be falsified.

So it's not really a hypothesis in this way, no?

But I was wondering, so there are these different variables that we have in this framework.

For instance, we have priors on hidden states.

So there are these different ways that could something like this be interpreted as a prime something that facilitates successive action versus, for instance, then we have habits or we have preferences, which might reflect more conscious kinds of decisions.

Would that make sense to think about the distinction this way?

Well, well, I'll go for it.

Yeah, just about the first point you mentioned, actually, it's FEP that because of its being a kind of principle is inherently unfalsifiable,

but active inference, at least as far as I understand it, is totally falsifiable theory.

So, for instance, we have some work done by Daniel, what was the name of the Japanese research?

Isamura, if I'm not mistaken, right?

Yes, Isamura.

Isamura, yeah.

Which have tried to find some neural correlates that for some of these active inference or some of the claims made within active inference framework.

And also there are other various studies that show the psychological evidence for those claims happening,

actually happening in the model, in the organism or agent of interest.

So that's about the first point. But I actually forgot the second point you make. So if you don't mind to reiterate it.

Yeah, so I was wondering whether those different kinds of variables that we have, for instance, priors on hidden states,

and then we have habits and then we have habits and all this. So in different places they sneak into this whole framework and whether they could be associated with those different kinds of cognitive processes.

So for instance, that we have priors, which condition somehow the sequence of states, whether this could be interpreted as a prime that facilitates certain kinds of activities of states that follow.

Whereas, for instance, or habits are more long term kinds of entrenched routines and that could be maybe reflect more conscious kinds of processes, something like this. Would it make sense to see it in this way?

I'll give a first thought. So first the falsification. So the principle cannot be falsified. And then active inference might be being deployed in a way where somebody is looking to explore whether it's adequate or not for a given setting.

Or it can merely be deployed where it's not even being tested against alternatives or evaluated as a framework itself, but merely being used to explain data.

The different variables and the interpretation of cognitive phenomena is the whole point of cognitive modeling.

So the prior in terms of it conditioning or initiating inferences on hidden states is what that variable does in the model.

However, there are a lot of open interpretations on like what is awareness and what is consciousness and all these other questions about timescale.

Like someone could argue that the slower things are not aware of.

But in the end, it's really going to come down to just writing the generative model and then it can be seen.

In figure 4.3, there's no consciousness, awareness, metacognition, etc, etc.

So the phenomena that this basal module provides are actually quite flexible, which is why PUMDPs are broadly used even outside of ActInf.

But they don't necessarily tackle the hard problem or anything like that.

So mapping it to what is experienced is certainly an area of open work.

But understanding how different cognitive phenomena arise in a model from the interplay of different parameters is what cognitive modeling is.

And there's very few attempts to try to relate this to this kind of psychological research or these kinds of psychological models.

It depends what you would mean by a lot.

Like, for example, association of precision or variance estimators with anxiety has been done in theory and in practice.

And association of different parametric evaluations with different cognitive states and pathologies is reviewed in Table 9.1.

So parameters are mapped to cognitive states.

That's why this kind of modeling is done.

But that doesn't mean that the framework has one singular way of interpreting it.

It's like you could make a linear model that does this or that, and you could test whether there's an effect of this condition on that outcome.

But you'd be talking then with your referent as the system of interest, not with your referent as somehow confirming or disconfirming something about linear models.

So that's why there are papers that just use active inference modeling to explain data.

Like everything that Ryan Smith is doing, everything that we see in Table 9.1.

But although sometimes those kinds of studies are taken as evidence for active inference, like look at these papers that use active inference empirically.

And they do support the notion that active inference can be used empirically.

But in these cases, there might be some, but I don't think most of them are this way.

They're not necessarily comparing continuous time generative models in ocular motor syndromes versus an alternative model.

Such a comparison would put it more clearly into the category of like unique explanation prediction value of active inference as opposed to other proposed, which is to say hypothesized cognitive models.

But without comparison of multiple methods in a paper, it's not evidence for that framework.

It just, it's evidence that that framework can be used.

And the mapping of cognitive phenomena to parameters is sometimes seems pretty clear.

Other times, especially if one is getting into more complex models, it's not going to be localized to like a single parameter because the cognitive phenomena will arise from different interacting parameters.

Like, well, when the learning rate on B is low, then this or that irregularity gets partitioned this way.

But when the learning rate on B is high, then this might happen in this environmental setting.

But again, there's so many conditioning factors that at the level of abstraction across all possible cognitive models, it's only right and good that active inference doesn't have such a one-to-one mapping between parameters and phenomena.

If there were such a mapping, we wouldn't have the expressivity to generate complex models.

And cognitive phenomena have many manifestations and many factors that influence them.

There are some promising avenues for that, but I'll keep it more focused on the textbook for here.

Okay, thank you.

Thank you.

With respect to this Marvin Minsky quote, in general, we are least aware of what our minds do best.

And which of these aspects are mapped onto experiential awareness?

Well, you could create multiple possible structural alternative cognitive models.

So I wonder if people are experiencing the O.

I wonder if they're experiencing the hidden state.

I wonder if they're experiencing B.

I wonder if, and then you could design the right experiment, or maybe the data already exists.

Where you show something and you say, now, what did you remember about that?

Or what was your experience while that was happening?

And you could find out.

And then you could evaluate and falsify one of those structural models.

Or you could take a more Bayesian approach and just describe the Bayes factor or the posterior model probability across those structural alternatives.

That quotation by Minsky also maps perfectly, I think, with Marc Solmes's claim on page 303 of his book, in Spring, which claims that briefly we only need consciousness or at least are aware of our consciousness when we're trying to correct for the more divergent divergence between our priors and our perceptions.

So, in other words, consciousness is something that's required for that correction or belief updating.

And as more precise our predictions become, the less we'll need our awareness of our consciousness.

Interesting. That makes me think about the flow state like St. Mahali and the kind of awareness associated with automatic performance.

But that's on the path towards empirical and experimental phenomenology.

Again, with all of the report biases and challenges associated.

And once you got to that point of specifying, just like figure 9.1 with the person and showing them different colors

and then the self-report and here's the input and output, it wouldn't be making claims about active inference overall, let alone the FEP.

Again, any more than some extensively developed multilinear model for healthcare data is making claims about linear modeling.

They're like kind of...

They're related.

But evidence of one is not the same type of evidence for the other.

But here they are going to abstract away from the specific active inference models and focus on integrative aspects.

So, now we're in the space of integrative models of cognition and behavior.

And so, with respect to that, if we merely see an area for improvements of active inference, great.

Then we're active inference researchers.

But it also makes sense to ask, what is active inference being compared with?

And because of the unifying scope being so vast,

it's hard to even conceptualize or name some alternative unified perspective.

But there are unifying things that might just be like even more general that end up being less helpful.

Like just saying, well, it's computation or it's information.

You know, memory is information.

Perception is information.

That might not be differentiated enough.

Whereas, hopefully, the positioning of active inference is, on one hand, unifying diverse cognitive phenomena in different systems.

But on the other hand, not so bland and generic that it's equivalent to just saying, you know, physics did it.

This book offers a systematic account of the theoretical underpinnings and practical implementations of

active inference.

Do you fellows agree with that?

Or how does that claim strike you?

Not so much practical, but systematic and theoretical, yes, I would think.

Thanks.

I broadly agree.

I agree.

I agree.

They're going to summarize the previous chapters.

Chapter one.

Organisms, or really just things, engage in adaptive exchange,

which from a cybernetics perspective, we can frame in terms of action and perception.

And they laid out the two roads.

low road we're using bayesian statistics

high road things must from the outside at least be re-measurable if it's not re-measurable

it doesn't have a stable identity from the outside to an observer that's like the low bar on the high

road if the volatility is so high that the thing is changing and not being what it is then it isn't

that and then the high bar on the high road is that actions must actually be taken to persist

and whether those actions are trivial or not is another question that's where we get into that

taxonomy of things with the path integrals paper but just broadly low road we're using bayesian

statistics is there really a modern alternative high road we're at least asking for repeated

measurements if not adaptive action for persistence again

uncontentious

chapter four approaches the generative models in active inference

exact bayesian inference may be implausible so they very rapidly move to variational inference as one

approach to approximate challenging bayesian statistical settings commonly used outside of

active inference contentious even the use of a variational free energy is contentious energy

based learning evidence lower bound now being applied to a comprehensive generative model that

includes perception cognition and action thus unifying for example signal processing on the inbound and control theory on the outbound

chapter five they went through three mammalian nervous system examples and gave a little bit of information on each of those systems in isolation

and then in the concluding figure wired them together to show how cortical inference could play a role in dopaminergic tone of the basal ganglia

which could then modulate the trade-off between reliance on habit

or conversely on expected free energy based updating of action priors into action posteriors

and then the ways that those selected action policies could then be enacted through a kind of simple differential mechanism in the spinal cord

from an epistemic perspective what else would you fellows have wanted to see in the first half of the book again considering that chapters one through five were like the theory and the underpinnings the theory and the underpinnings and seeing this as just the first version of what it can be epistemically what else might be important to include in these chapters whether it's just a micro edition like oh they could have added this citation or this variable or whether you have a more structural comment ollie go for it

ollie go for it

i'm not sure if i've uh said this before uh or just um talked about it uh casually but uh in my opinion some of the materials from chapter 10 uh could be moved to some of the earlier chapters in order to uh give a kind of bigger picture uh in which um active inference fits uh so uh

reading the first chapters uh the reader at least uh in my view doesn't get so much um

i mean doesn't learn about the historical context of the ideas uh as as much as needed because uh frankly enough

reader

uh uh i suppose will not be uh familiar too much with uh active inference framework because uh obviously it's a relatively uh new framework so uh it might help them to see uh the distinction between active inference and uh the other uh earlier uh frameworks such as bayesian brain hypothesis predictive coding predictive processing and so on um and uh yeah just

uh see how active inference

it fits uh within that bigger

uh

yes thank you ali what do you think about reading chapter 10 first

outlining all the chapters that would be ideal i guess

yeah exactly wow unbelievable

we could try it we could try it we could try it michael

yeah we cannot we cannot really know because we did it the other way around

so but you can maybe try it in the next cohort not to see how that works out for you

yeah that would be a funny frame like we're gonna read chapter 10 so truly do not worry about the earlier references they might mention an earlier chapter again you don't have to worry because you won't have read it but this is where we're going now you've been there that's the big picture and now we're gonna work forwards via backwards to get to that integrative um conclusion and then we can ask whether that

um was reached was that where we wanted to go and did we reach it with integrity but it's not going to be a math test

i think that makes sense to go that way

yeah

interesting

because this is so epistemic and 10 is a relatively long chapter with no equations no figures

and so in that way it's like a super version of chapter one
including bringing up many of the same ideas
so definitely for our consideration in the coming weeks
okay in chapter six there was a this is the pragmatic section of the book allegedly
chapter six
chapter six provided a recipe
i'm sure we had many questions about what else was happening at that restaurant and a little bit more detail
from the sous chef and all of that
but a recipe was provided which we can continue to build out
and they just buzz through chapter seven eight nine in review
merely saying that seven and eight focused on discrete and continuous time generative models respectively
and chapter nine opens the door to computational phenotyping
in terms of the framing and opportunities and challenges associated with modeling empirical data
and uses of generative models for statistical ends
so
more review on the first five
then a little bit on six speeds are seven and eight
eight through seven and eight
self-evidencing as a first principle
not reward as a first principle
self-evidencing relative to some active generative model as a first principle
if you're on board
enjoy the ride
go back to chapter one read the book
if you're not convinced about self-evidencing
as a first principle then there might be some other approaches still go back to chapter one and read it but
once somebody is on board
that perception and action being in the same game
of self-evidencing
with a common information theoretic currency
information related currency
active inference is that model
we do more than integrate perception and action
we don't homogenize them or blend them
it's actually very clear what they are
but they're in the same game and they have that same currency
and when we take that kind of a first principles approach
it's compatible with these kinds of inactivist

languages

but also

have

extensible ways to deal with other diverse cognitive phenomena

like memory and attention

how do we deal with all these cognitive phenomena

in different systems with different architectures and so on

well

everything

is reflected through structural or parametric

specification in the generative model

part one

part two

generative models are evaluated according to their free energies

just like any other bayesian model

part two

so part one

part two

say what you mean

in the generative model

if it's not in the generative model

it's just a secret in your mind

part two

if it's a bayesian graph generative model

then we have techniques for model fitting

message passing

variational free energy

and so on

step one

step one

step one

don't make it a secret

step two

fit the model

like all models can be fit

super clear

super powerful

sometimes hard to even convey
because it's so
removed
from the presentation
of other cognitive science approaches
earlier

shank
and the neat scruffy distinction
was introduced
chapter one probably
active inference is not
reifying
the nouns in english
associated with cognitive functions
and trying to cobble them together
together
it's not a mosaic
it starts
from a unified
perspective
and then
elaborates or differentiates
in order to realize
cognitive functions
but not by reifying them as modular items
that can be like picked up and moved
again
really key
different than
how it's often addressed
in other areas
probably a lot of philosophical
depth that could be explored
not for today
in these jamesian categories
when we take that perspective
of starting with a unified

whole iguana

a whole conceptual iguana

whole

behavioral ethologist

looking at an iguana

situation

figure 9.1

then

we can understand

so much better

structural and parametric

similarities and differences

amongst different cognitive models

we couldn't ask for a better way to do it

if one appears

it'll be a gift

but within

some pretty

broad considerations

this

ethologist's

setup

is the one that we want

for this making a map

about a map

and that's why

um

saketeve

devel

ramstead and friston

are such vociferous advocates

for everything that we're seeing

the trade-offs

amongst all kinds of behavior

one example being like

so-called explore exploit

but all kinds of cognitive trade-offs

like forgetting versus
attention and
risk versus
etc
all of those
when
approached as if they could be
these modular standalone
patterns
they are only reified
through investigation
rather than
seeing these trade-offs
these pareto-optimal
trade-offs
or these manifolds
of measured
outcomes
rather than
looking at the generative model
or the generative process
that gives rise to those
manifolds
the manifold becomes
reified as if there was
somehow a fundamental
trade-off between
exploration and exploitation
when
depending on the
regularities of the
niche and the affordances
available to the agent
those two might literally
not be at odds
like what if you're just
munching through a field
and there's um

food everywhere
and you're just munching
all around
then there's no trade-off
so the trade-off is not
fundamental to these
english words
it's about the interplay
of the generative model
and the generative process
which is our stance
on everything
it's about the interplay
between the generative model
as specified
and the generative process
as specified
trade-offs don't exist
in some other
abstract space
beyond helping us
at the
algorithmic
layer here using
mars taxonomy
we can also
peek deeper
into neural computations
and representations
signatures
and so on
and here
one can
look at the
work
in spm
in human
neuroimaging

and sensor fusion
and then
build from there
to incorporate
more sophisticated
generative models
and action policies
now they're going to
sketch a
psychology textbook
and this is where
the earlier point
about 10
providing a lot
of the conceptual
and historical context
is very
salient
because
these are the kinds
of things
these are the
the buoys
or the signposts
that other people
are going to be familiar
with
when they're wanting
to know more
about
active
they introduce
and discard
the sandwich
model
and
they
highlight

the primacy
or the centrality
of the generative
model
that explicitly
makes predictive
and
arguably
goal-directed
but at the very
least predictive
dimensions
of that model
absolutely
primary
what imperative
is downstream
of the primacy
of prediction
self-evidencing
at the stimuli
processing level
we see predictive
coding
predictive processing
at a slightly slower
which we
timescale
which we could
model with a
nested model
we see learning
neuromodulators
!
slower inference
and
action is
fundamentally

embedded within
this process
for example
eye saccades
moving across
the page
this
allows
the dual
drives
which epistemic
and pragmatic
are one
decomposition
of
free energies
expected free
energy in this
case
to integrate
the sort of
preference oriented
goal realizing
and
information oriented
components to
behavior
predictive
processing
what is there
to say
this is basically
predictive processing
plus action
and live stream 43
makes that
pretty clear
the paper in

live stream 43

perception

you don't see

with your eyes

you see with your

mind

what you're

seeing is

generated

you're not

getting the

video camera

raw feed

that's a

classic idea

!

Maria did

great work

reporting on

the history of

these ideas

in 43.0

bayesian brain

they returned

to predictive

coding

action

action is

another variable

we make

predictions about

action

we have priors on

action

we have a policy

prior

and then we

update it or not
into the policy
posterior
that's
that basal
ganglia
chapter 5
so
action
selection
planning
as
inference
is just
another variable
and so
all of the
meta
on bayesian
statistics
is going to
hold whether
we're talking
about
inference
on perception
the kind
of
under
part
of figure
4.3
or
inference
on action
the upper
part
of figure

4.3

within each
of the
GMs
and that
leads to
some different
thinking
about
for example
motor
control
and
interactionism
versus
instructionism
sensory
attenuation
and all
these other
diverse
motor
phenomena
also
being
integrated
because
we took
a first
principles
perspective
we didn't
choose
to
reify
sensory
attenuation
as some

discrete
phenomena
rather we
chose to
model
action
as a
kind
of
inference
action
selection
as a
kind
of
inference
that an
embodied
agent
takes
in the
niche
and that
led to
a complete
collusion
with sensory
attenuation
as an
empirical
phenomena
this
isn't
falling
out of
reward
learning
this

isn't
falling
out of
excel
spreadsheets
idea
motor
theory
and various
other
paradigms
have been
proposed
to understand
this
bi-directionality
with action
and effect
so it's
like a
little bit
jarring
to read
because it's
kind of
not in the
active
inference
ontology
how it's
being stated
by them
of course
but
many of
the terms
are
resemblant

enough
and
active
inference
encompasses
it
and bolsters
it
and provides
ways to
compute
message
passing
all of
that
anything
that's
just
like a
way
of
thinking
it's
like
that's
cool
it's
a cool
way
of
thinking
can
you
do
message
passing
on
that

so
that's
where
the
pragmatic
value
comes
into
play
there's
a variety
of ways
of thinking
about the
action
perception
loop
there's
the
tote
test
operate
test
exit
there's
the
OODA
observe
orient
decide
act
and so
on
and
broadly
we're
talking about
the same

kinds
of
agents
and
active
inference
helps
us
understand
frameworks
that have
often
been
qualitatively
applied
or
quantitatively
applied
to
cybernetic
systems
on
the
inbound
optimal
perceptual
processing
is a
signal
processing
challenge
on
the
outbound
optimal
action
selection
has been

approached
from
this
area
of
control
theory
we're
unifying
the
inbound
signal
processing
and
the
outbound
control
theoretic
dimensions
of
cybernetics
into a
unified
imperative
how
exactly
they're
balanced
in a
given
realization
of a
model
that is
for the
researcher
to specify
not for

the framework

to be

opinionated

upon

but

we have

the

grammar

and

the

space

to

have

blind

systems

that

act

wisely

or

not

and

perceptive

systems

that

act

intelligently

or

not

or

whatever

the

framework

is not

going to

have

an

opinion

on

those
kinds
of
balances
in
their
specifics
special
cases
of
the
unified
imperative
for
example
the
expected
free
energy
when
there
is
no
epistemic
value
to
gain
fully
observable
setting
perfect
understanding
of the
consequences
of the
self
in the
world

or on
the
other
hand
a
setting
where
there's
a lot
to
learn
but
there's
no
pragmatic
value
all
outcomes
are
equally
preferred
special
cases
of
decision
making
map
onto
special
cases
of
the
unified
imperative
so
just
from a
very

narrow
and
technical
view
active
inference
is
simply
a
generalization
of
for
example
Bayesian
decision
theory
reinforcement
learning
but let's
see how
they differ
they
differ in
a usage
of a few
ontology
terms
they
talk
about
model
based
and
model
free
RL
policy
gradient

methods
lot
of
similarities
of course
it'd be
awesome
to have
a notebook
where we
can
compare
and
contrast
side
by
side
action
planning
as
inference
it's
just
another
variable
and
other
people
have
suggested
it
approximate
Bayesian
computation
bounded
rationality
base
optimal

behavior
including
pathology
these are
all
totally
plausible
for us
to explore
valence
emotion
motivation
these are
quite
diffuse
and complex
cognitive
phenomena
there are
ways to
specify
them
in
generative
models
doesn't
mean
that
actin
tells us
what these
things are
they're
just
strings
of
english
letters

they're
just
pixels
on
a
screen
but
we
can
use
them
semiotically
and build
cognitive
models
that have
these
kinds
of
patterns
homeostasis
of course
considered
broadly
is critical
to the
high road
and so
we have
a lot
about
how
intero
and
extero
septive
inference
interact

in order
to support
persistence
attention
and salience
relevance
realization
these are
core for
the epistemic
drive
homeostasis
can broadly
be seen
as being
about
pragmatic
value
whereas
epistemic
dynamics
are about
epistemic
value
different
cognitive
modes
again
not just
being
proposed
ad hoc
by a
TED talk
or a
pop
psychology
book

or a
research
paper
starting
from a
solid
grounding
and then
differentiating
the
generalized
framework
to embody
these
patterns
not to
reify
the patterns
as actual
phenomena
that then
need to
be plugged
and played
and we
know there
are so
many open
directions
applications
to human
scale
social
cultural
systems
integration
with
contemporary

machine
learning
systems
embodied
cyber
physical
systems
sometimes
called
robotics
and
just like
the cover
of the
book
was the
one ring
not to
rule
them all
here we
have the
Tolkien
quote
and
the
journey
comes to
a
conclusion
unless you
read on
to the
appendices
they
proposed
an
introduction

and
at least
from the
author's
view
you will
pursue
active
inference
in some
form
so
in our
closing
minutes
how does
this
strike
either of
you fellows
what are
your reflections
upon completing
the book
one or
more times
yeah i'm
very much
convinced about
this idea
and i guess
i will
pursue this
for some
time to
come
to
lots of

work to
fit the
data into
this whole
thing
but that's
my
intention
and thanks
very much
for your
very insightful
guidance
through this
book
it was
extremely
helpful
thank you
thank you
so much
thank you
michael
ali
upon this
nth reading
of the
textbook
what is
standing out
to you
more and
more or
differently
in fact
more generally
i feel
that with

each
reading
a kind
of nested
markup
blanket
is building
up
inside
my
i mean
cognition
framework
because
with each
reading
i become
much aware
of some
of the
connections
between
different
parts
and different
sections
of the
theory
which i was
not aware
on the
first reading
at all
so hopefully
this growing
nested
markup
blanket

will continue
to grow
with
future
reading
as well
thank you
awesome
yes
what to
even
say
it was a
great
cohort
we spanned
in the end
of 2022
on through
april
and 23
um
and that
brings this
cohort to
a conclusion
so
thanks a lot
to
cohort two
next week
we'll return
and it won't
be recorded
we'll just be
talking about
feedback
project ideas

and so

on

and um

thanks again

from